

Module 03

Docker Basic Commands

DevOps end to end services & solutions

Agenda

- > Docker – Basic Commands
- > Docker run
- > Docker ps, Docker ps -a
- > Docker stop, start and restart Commands
- > Docker rm
- > Docker logs, Docker exe -it
- > Docker tag and push to Registry
- > Hands-On Lab-02 MySql and Lab-03 Jenkins

Basic Docker Commands



docker version

Check the installed
Docker version.



docker info

Display system-
wide information.



docker help

Get help with
commands.

Docker Advance commands based on httpbin app

Runs a new container. If the image is not available locally, Docker pulls it from the registry.

```
docker run --name http_error -d -p 8990:80kennethreitz/httpbin
```

- > **-d**: Detached mode (background)
- > **-p 8990:80**: Redirect host port 8990 to container port 80
- > **--name**: Assigns a custom name

Container Management

- > **docker run:** Create and start a container
- > **docker ps:** List running containers
- > **docker ps -a:** List all containers
- > **docker stop:** Stop a running container
- > **docker start:** Start a stopped container
- > **docker restart:** Restart a container
- > **docker rm:** Remove a container
- > **docker logs:** View logs of a container

Let's explore these in detail.

Container Management

- > **docker top:** It shows running processes inside a container
- > **docker stats:** List running Shows live resource usage of containers in real-time
- > **docker container inspect:** List all containers It shows full detailed information about a container in JSON format
- > **docker image inspect:** It shows **detailed metadata about a Docker image** in JSON

Let's explore these in detail.

Container Management

- > **docker commit:** it creates a new image from a running (or **stopped**) container
- > **docker save:** It saves a Docker image as a .tar archive
- > **docker cp:** It copies files/folders:
- > **docker tag:** give your image a registry-ready name
- > **docker push:** upload that image to a registry (like Docker Hub)

Let's explore these in detail.

Docker ps

Lists containers that are currently **running**.

```
docker ps
```

Key Columns:

- **CONTAINER ID:** Unique identifier
- **IMAGE:** Image used
- **STATUS:** Up / Running time
- **PORTS:** Port mappings (e.g.,
0.0.0.0:8990->80/tcp)

CONTAINER ID	IMAGE	STATUS	NAMES
a1b2c3d4e5	httpbin	Up 2 mins	http_error

Docker ps -a

Lists **all** containers, including those that are running, stopped, or exited.

```
docker ps -a
```

CONTAINER ID	IMAGE	STATUS	NAMES
2ffdd2cbb26c	final-project	Exited (137)	final-project
Exited (0)			

Useful for finding stopped containers to restart or remove.

Lifecycle Commands



docker stop

Gracefully stops a running container.

```
docker stop
```



docker start

Starts a container that was previously stopped.

```
docker start
```



docker restart

Stops and then starts a container in one step.

```
docker restart
```

Docker Build and Run

Docker top < C ID >

It shows you **what processes are actually running inside a container**

Docker stats < C ID >

live performance dashboard for containers
It shows **real-time resource usage** — CPU, memory, network, and disk I/O

Docker Build and Run

Docker inspect < C ID >

deep-dive X-ray of Docker

It shows **everything Docker knows** about an image or container — config, networking, mounts, env vars, ports, PID, etc

Docker logs –follow < C ID >

watch a container's output live

Like less

Docker logs < C ID > | less

Docker rm

Removes one or more **stopped** containers.

Bulk Removal:

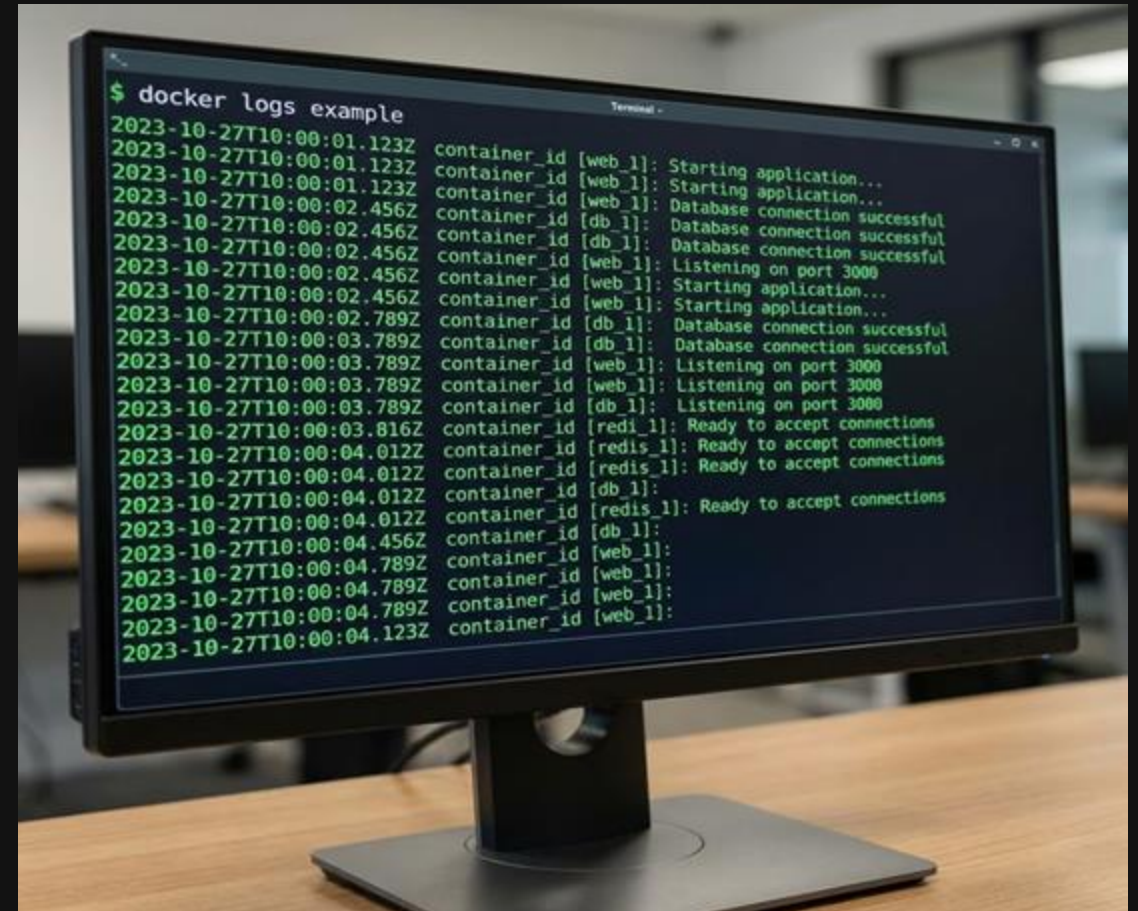
```
# Basic removal  
docker rm container_id  
  
# Force remove running container  
docker rm -f container_id
```

```
# Remove all stopped containers  
docker rm $(docker ps -aq)  
  
# Remove filtered containers  
docker rm $(docker ps -q -f status=exited)
```

Docker logs

Retrieves logs from a container.

- Show only last 100 lines.
- Follow log output in real-time.
- **docker logs -f --tail 100 mongo**



Docker exec

Interactive Session

Run a command inside a running container.

```
docker exec -it mongo bash
```

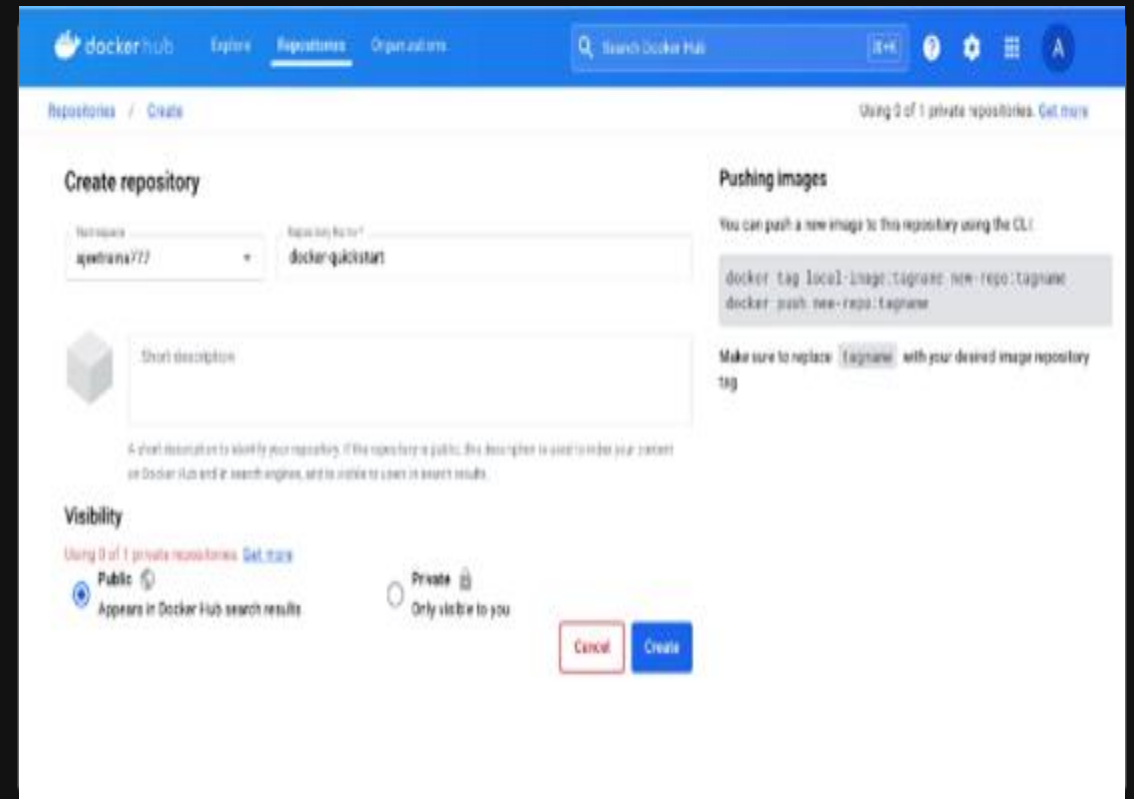
Understanding Flags

- i:** Interactive. Keeps STDIN open.
- t:** TTY. Allocates a pseudo-terminal.

Docker Registry – Tag and Push

Steps to Publish

- > **Login:** Authenticate with Docker Hub.
- > **Tag:** Rename your local image to match your registry repository.
- > **Push:** Upload the image to the registry.



The screenshot shows the Docker Hub 'Create repository' page. The header includes the Docker Hub logo, navigation links for 'Explore', 'Repositories', and 'Organizations', a search bar, and user account controls. The main content area is titled 'Create repository' and includes a 'New repository' dropdown set to 'yehuda777', a 'Repository name' field containing 'docker-quickstart', and a 'Short description' text area. To the right, the 'Pushing images' section provides CLI commands: `docker tag local-image:tagname new-repo:tagname` and `docker push new-repo:tagname`. Below this, the 'Visibility' section shows 'Using 0 of 1 private repositories' and two radio buttons for 'Public' (selected) and 'Private'. The 'Public' option is described as 'Appears in Docker Hub search results'. The 'Private' option is described as 'Only visible to you'. At the bottom right are 'Cancel' and 'Create' buttons.

Tag & Push Commands

1. Login to Docker Hub

docker login

2. Tag the image

docker tag /:

docker tag myapp mydockeruser/myapp:latest

3. Push the image

docker push mydockeruser/myapp:latest

4. Pull (Verify)

docker pull mydockeruser/myapp:latest

Hands-On – Lab 02: MySQL

- Docker pull mysql
- Docker run with port 3406
- List of all images
- Exec into running mysql container
- Run SQL: show tables;



Lab 02: Cleanup



Stop

Stop MySQL container



Start

Start MySQL container



List

List all
containers



Remove

Remove
container

Hands-On – Lab 03: Jenkins

- > Run Jenkins (pull & run in one command)
- > Open Jenkins via browser
- > Create "Hello world" job
- > View Jenkins logs



Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

